

Trigonometric Equations (Part II) Homework

Name
Date
Course

* When solving the following problems, do not use a calculator if your answer(s) are 0° , multiples of 30° , or multiples of 45° .

① Solve if $0 \leq W \leq 360^\circ$.

$$-3\sec W - 2 = \tan^2 W + 1$$

② Solve if $0 \leq A < 360^\circ$.

$$4\tan A + 2 = 6\cot A$$

③ Solve if $0 < \theta \leq 360^\circ$

★ Hint: There is no solution. But, you must show why there is no solution, as we did in class.

$$\sin \theta + 3 \csc \theta = \cot \theta$$

④ Solve if $0 < A < 360^\circ$

$$\sin(2A) = \frac{1}{2}$$

⑤ Solve if $0 < \alpha < 360^\circ$

$$\cot\left(\frac{\alpha}{2}\right) = -\frac{1}{\sqrt{3}}$$

⑥ Solve if $0 \leq B \leq 360^\circ$.

$$\sec(3B) = \sqrt{2}$$

⑦ Solve if $0 \leq \theta < 360^\circ$.

$$2 + 25 \tan(2\theta) = 2 + 24 \tan(2\theta)$$

⑧ Solve if $0 \leq W \leq 360^\circ$.

$$-3 \sec\left(\frac{W}{2}\right) + 4 = 1$$

⑨ Solve if $0 \leq \beta \leq 360^\circ$.

$$\csc(3\beta - 180^\circ) = -2$$

⑩ Solve if $0 < \alpha \leq 360^\circ$.

$$\cos\left(\frac{\alpha}{2} + 90^\circ\right) = 0$$

Homework Answers

① $W = 120^\circ, 180^\circ, 240^\circ$

② $A = 45^\circ, 123.69^\circ, 225^\circ, 303.69^\circ$

③ No solution $\cos \theta \neq \frac{-1 \pm \sqrt{17}}{2}$

④ $A = 15^\circ, 75^\circ, 195^\circ, 255^\circ$

⑤ $\alpha = 240^\circ$

⑥ $B = 15^\circ, 105^\circ, 135^\circ, 225^\circ, 255^\circ, 345^\circ$

⑦ $\theta = 0^\circ, 90^\circ, 180^\circ, 270^\circ$

⑧ $W = 0^\circ$

⑨ $\beta = 10^\circ, 50^\circ, 130^\circ, 170^\circ, 250^\circ, 290^\circ$

⑩ $\alpha = 360^\circ$